



DEALER BEST PRACTICE SERIES

Failure Analysis Report

Maintenance and Repair Application Component Life Management Component Renewal (CRC) MARC Management

Failure Analysis Report.....0

1.0 Introduction.....1

2.0 Best Practice Description.....1

3.0 Implementation Steps.....9

4.0 Benefits.....9

5.0 Resources Required.....9

6.0 Supporting Attachments / References.....9

7.0 Related Best Practices.....9

8.0 Acknowledgements.....9

DISCLAIMER: The information and potential benefits included in this document are based upon information provided by one or more Cat® dealers, and such dealer(s) opinion of “Best Practices”. Caterpillar makes no representation or warranty about the information contained in this document or the products referenced herein. Caterpillar welcomes additional “Best Practice” recommendations from our dealer network.

1.0 Introduction

Trakindo is one of the largest dealerships in the world, due to the nature of the archipelagos they have over 60 branches and 20 site offices that make up their dealership network. Each one of the branches has specific capabilities from parts distribution, field service capability to component rebuilds. From these outlets they have put together a series of best practices that they have recognized internally to help all the branches.

This best practice is dealing with Failure Analysis Management. It begins with the complete report to the specially supporting document for warranty, any redo reports and finally product problems.

The reason for this best practice is that Failure Analysis Management has a big impact on Trakindo's working business. After carrying out surveys at their major branches Trakindo has found that there was no consistency in a standard process and document to ensure all the complete documentation is available at time reporting for Failure Analysis.

2.0 Best Practice Description

Previously Trakindo had no standard documents for Failure Analysis reporting. The documents that was available were incomplete in content.

This new standard document will ensure that the technician or respective person can fill out all the correct available information and follow up with the respective person if any information is missing. Long term it will also standardize the Failure Analysis Report management and ensure that with cross transfer of people they are using the same process across all branches.

This best practice will also save time in reporting Failure Analysis, hopefully speeding up warranty claims and carry out in solving problems that occur.

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Failure Analysis Report	DATE 1/13/2015	CHG NO 01	NUMBER 0911-1.04-1226
-------------------------	-------------------	-----------------	--------------------------

Below is the old format for managing Failure Analysis reporting process.

COMPONENT FAILURE ANALYSIS REPORT					
CUSTOMER		LOCATION		DATE	27 February 2009
MACHINE MODEL	777C	SERIAL NO		UNIT NO	4111
ENGINE MODEL	3508	SERIAL NO	-	Serial	93131
DEALER	PTTU	CODE		LOCATION	
WORK ORDER NO		WO NO			
COMPONENT	Upper Starting motor				
COMPONENT LIFE	7449 hrs	INSTALL DATE	22 Nov 2007	Serial	75662 hrs
COMPONENT LIFE	27 months	FAILURE DATE	26 Feb 2009	Serial	93131 hrs
GROUP NO:	1W5363	DESCRIPTION	Starting motor gp electric		
PART NO:	10R1914	DESCRIPTION	Starting motor		
REPORTED BY	Juanita	SALARY NO			
APPROVED BY					

Trakindo CAT		FAILURE ANALYSIS REPORT	
Form Title	RKCRG Failure Analysis Report	Replacement WO	1
Form No	OF SS 0004-04	Date Rebuilt	1
Rev / Date	04 / 01 July 2006	FA WO	1
Authorized	SMR	PTTU WO	1
		Document Name	1
DEALER CODE	J367	WORK ORDER	
DATE	05 / 07 / 2009	EMPLOYEE IDENTIFICATION	
CATERPILLAR MODEL	D11R	CATERPILLAR SERIAL No	7PZ01184
HOURS	7614	KILOMETERS	
CUSTOMER NAME			
EQUIPMENT LOCATION			
INSTRUCTION	Rebuild Engine		
PART NUMBER RESPONSIBLE	PART NAME	QTY	DESC CODE
7N-3006	Thrust washer	1	10
GROUP NUMBER CONTAINING PART	GROUP NAME	DO THIS MODERN MAKE THE PRODUCT INOPERABLE	DESCRIPTION OF COMPONENT USE (SPACE MAXIMUM FOR INCIDENT)
8N-8161	Drive Gp Accessory	☐ YES ☐ NO	Broken
DESCRIPTION CODES * A - Structural C - Leaks E - System Malfunction G - General Repair K - Serviceability S - Operator B - Hydraulic D - Factory Assembly F - Factory Shipping H - Adjustment N - Abuse Component			
Equip Number	DZ-170	PCR Schedule Hrs	12000 hrs
Component	Engine Gp	Comp Hours	7614 hrs
Comp Serial No		Life Remaining	4386 hrs
Mach Hrs Install	13010 hrs	Cost To Repair	
Mach Hrs Removed	20624 hrs	Cost/Hr Remaining	\$ /hr
Date Comp Installed	15 Feb 2008		
Date Comp Removed	03 Apr 2009		
Report By / Date	Bayu M/08-05-09		
Comp New Price		Sgt Checked by	Supt Approved by

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Failure Analysis Report

DATE
1/13/2015

CHG
NO
01

NUMBER
0911-1.04-1226

	FAILURE ANALYSIS REPORT
---	---

Trakindo WO : _____	Reported By : _____
Original WO : _____	Reported Date : _____
Checked by Foreman / Supervisor : _____	Approved by Service Head : _____
(Name and Signature)	(Name and Signature)

SIMS Information

DEALER CODE : _____	EQUIPMENT NUMBER : _____	CUSTOMER NAME : _____
DATE : ____ / ____ / ____ (Mo / Da / Yr)	TECHNICIAN IDENTIFICATION : _____	EQUIPMENT LOCATION : _____
CATERPILLAR MODEL : _____	CATERPILLAR SERIAL NO : _____	INSTRUCTION : _____
HOURS : _____	KILOMETERS : _____	

PART NUMBER RESPONSIBLE	PART NAME	QTY	DESC CODE	GROUP NUMBER CONTAINING PART	GROUP NAME	DID THIS INCIDENT MAKE THE PRODUCT INOPERABLE	DESCRIPTIVE COMMENTS (3 SPACES MAXIMUM PER INCIDENT)
						☐ Yes ☐ No	
						☐ Yes ☐ No	
						☐ Yes ☐ No	

DESCRIPTION CODES :
(Refer SIMS Guidelines : SEB00350)
A - Structure
S - Surface
C - Leaks
D - Factory Assembly
E - System Misfunction
F - Factory Shipping
G - General Repair
H - Adjustment
W - Serviceability
N - Abuse
X - Operator
Complaints

Component Information

Component Description : _____	PCR Schedule Hrs : _____	Comp Target hrs (ex: 12000)
Comp Serial No / ID : _____	Comp Hours : _____	Current hrs failed (ex: 8000)
Comp Am / Gp No : _____	Life Remaining : _____	Remain of target life (4000)
Mach Hrs Install : _____	Comp New Price : _____	
Mach Hrs Removed : _____	Cost To Repair : _____	(ex: \$ 10000)
Date Comp Installed : _____	Cost/Hr Remaining : _____	(10000 / 4000 = 2.5 \$ /hr)
Date Comp Failed : _____		

Supporting Documentation Required

#	Description	Y/N	Comments
1	Installation Check List		
2	Repair & Maintenance History		
3	SOS History		
4	Magnetic Plug Inspections		
5	ET/VIMS Data		
6	TPMS / Payload Data		
7	Performance Test Data		
8	Others		

Failure Outline: Brief outline of what the Customer reported.

Background / Research: Include all observations made during your Failure Analysis investigation including good and bad.

Bagian dari / Part of : TSO-TEC-SOP-001-R00
1 of 2
Form Number : PSDV-SVTC-FRM-001-R00

	FAILURE ANALYSIS REPORT																																					
<table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="width: 50%; text-align: center;">Part Number -Name - Modifier Code (RH/LH/Upper/Lower etc)</th> <th style="width: 50%; text-align: center;">OBSERVATION To Help Explain Progression/Stage of Failure, Use - Notes /Photos/Parts Book Illustration. Identify Type, Amount, Location of Wear, Deposits, Damage, Fractures etc.</th> </tr> </thead> <tbody> <tr><td> </td><td> </td></tr> <tr><td> </td><td> </td></tr> <tr><td> </td><td> </td></tr> <tr><td> </td><td> </td></tr> <tr><td> </td><td> </td></tr> <tr><td> </td><td> </td></tr> <tr><td> </td><td> </td></tr> <tr><td> </td><td> </td></tr> <tr><td> </td><td> </td></tr> <tr><td> </td><td> </td></tr> <tr><td> </td><td> </td></tr> <tr><td> </td><td> </td></tr> <tr><td> </td><td> </td></tr> <tr><td> </td><td> </td></tr> <tr><td> </td><td> </td></tr> <tr><td> </td><td> </td></tr> </tbody> </table>	Part Number -Name - Modifier Code (RH/LH/Upper/Lower etc)	OBSERVATION To Help Explain Progression/Stage of Failure, Use - Notes /Photos/Parts Book Illustration. Identify Type, Amount, Location of Wear, Deposits, Damage, Fractures etc.																																	Conclusions (What Caused the Failure): Describe the root cause and the sequence of events that caused it. Warranty Decision or Recommendation: Please mark X in only one box below. <table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 33%; text-align: center;">Not warranty : ☐</td> <td style="width: 33%; text-align: center;">Warranty Accepted : ☐</td> <td style="width: 33%; text-align: center;">ASA Claim* : ☐</td> </tr> </table> Comments: *) Application Severity Analysis Solution to Prevent Reoccurrence: Check SIS for Service Letters and Service Magazines related to this failure. AMT history, MIM's history, Oil sample results. Please attach supporting documents below:	Not warranty : ☐	Warranty Accepted : ☐	ASA Claim* : ☐
Part Number -Name - Modifier Code (RH/LH/Upper/Lower etc)	OBSERVATION To Help Explain Progression/Stage of Failure, Use - Notes /Photos/Parts Book Illustration. Identify Type, Amount, Location of Wear, Deposits, Damage, Fractures etc.																																					
Not warranty : ☐	Warranty Accepted : ☐	ASA Claim* : ☐																																				
Baglaniden / Part of : TSD TEC.SOP.001 R00	2 of 2	Form Number : PSOV.SVTC.FRM.001 R00																																				

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Failure Analysis Report

DATE
1/13/2015

CHG
NO
01

NUMBER
0911-1.04-1226

Sample Report

		FAILURE ANALYSIS REPORT	
Trakindo WO : XX 61124		Reported By :	
Original WO : XXXXXX		Reported Date : 7 May 2009	
Checked by Foreman / Supervisor:		Approved by Service Head:	
(Name and Signature)		(Name and Signature)	

SIMS Information			
DEALER CODE	J XXXX	EQUIPMENT NUMBER	XX 61124
DATE	05 / 07 / 09 Mo / da / yr	TECHNICIAN IDENTIFICATION	
CATERPILLAR MODEL	D11R	CATERPILLAR SERIAL No	7PZ01164
HOURS	7614	KILOMETERS	-
		INSTRUCTION Rebuild Engine	

PART NUMBER RESPONSIBLE	PART NAME	QTY	DESC. CODE	GROUP NUMBER CONTAINING PART	GROUP NAME	DID THIS INCIDENT MAKE THE PRODUCT UNOPERABLE	DESCRIPTIVE COMMENTS (20 SPACES MAXIMUM PER INCIDENT)
7N-3006	Thrust washer	1	10	8N-8161	Drive Gp Accessory	☐ Yes ☐ No	Broken
						☐ Yes ☐ No	
						☐ Yes ☐ No	

DESCRIPTION CODES: (Refer SIMS Guidelines: SEB00850)	A - Structural B - Surface	C - Leaks D - Factory Assembly	E - System Malfunction F - Factory Shipping	G - General Repair H - Adjustment	K - Serviceability N - Abuse	X - Operator Complaint
---	-------------------------------	-----------------------------------	--	--------------------------------------	---------------------------------	------------------------

Component Information			
Component Description :	Engine Grp.	PCR Schedule Hrs :	12,000
Comp Serial No / ID :	TU-XX	Comp Hours :	7614
Comp Am / Gp No. :	1000	Life Remaining :	4386
Mach Hrs Install :	13010	Comp New Price :	\$...
Mach Hrs Removed :	20624	Cost To Repair :	\$... (5x \$ 10000)
Date Comp Installed :	15 Feb 2008	Cost/Hr Remaining :	(10000 / 4000 = 2.5 \$ /hr)
Date Comp Failed :	03 Apr 2009		

Supporting Documentation Required			
#	Description	Y/N	Comments
1	Installation Check List	Y	
2	Repair & Maintenance History	Y	
3	SOS History	Y	attached
4	Magnetic Plug Inspections	Y	attached
5	ET/VIMS Data	Y	attached
6	TPMS / Payload Data	Y	attached
7	Performance Test Data	Y	attached
8	Others		

Bagian dari / Part of : TSD, TEC, SOP, DO, R00	1 of 4	Form Number : PSDV, SVTC, FRM, DO, R00
--	--------	--

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Failure Analysis Report

DATE
1/13/2015CHG
NO
01NUMBER
0911-1.04-1226

 FAILURE ANALYSIS REPORT	
Failure Outline: Brief outline of what the Customer reported. Found much cooper debris at oil filter	
Background / Research: Include all observations made during your Failure Analysis investigation including good and bad. During PM found much copper debris in oil filter. Customer decided to remove the engine and sent to KKCRC for recondition	
Part Number -Name – Modifier Code (RH/LH/Upper/Lower etc)	OBSERVATION To Help Explain Progression/Stage of Failure, Use – Notes /Photos/Parts Book Illustration. Identify Type, Amount, Location of Wear, Deposits, Damage, Fractures etc.
Engine condition during received at KKCRC. 	
Oil pan after been removed. We found two pieces of thrust washer inside this oil pan which later identified coming from rear idler gear for PTO	
Bagian dari / Part of : TSD.TEC.SOP.001.R00 2 of 4 Form Number : PSDV.SVTC.FRM.001.R00	

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Failure Analysis Report

 DATE
1/13/2015

 CHG
NO
01

 NUMBER
0911-1.04-1226



FAILURE ANALYSIS REPORT

The opposite thrust washer toward the front of engine still attach on the flywheel housing. This mean the broken thrust washer is on the side toward the rear of the engine	
Complete picture of above thrust washer. The face toward the flywheel housing is worn out as deep groove due to rotating and rubbed by the dowel. While the face toward the gear has deep worn out as in friction with the gear. These happened after the opposite thrust washer broken and drop down causing increase of clearance/end play	
Shaft idler. The face of the shaft against the thrust washer found in normal condition	
The idler gear also found in normal condition. The face against the thrust washer found a bit worn due to friction with the thrust after broken.	

Baglan dari / Part of : TSD.TEC.SOP.001.R00

3 of 4

Form Number : PSDV.SVTC.FRM.001.R00

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Failure Analysis Report

DATE
1/13/2015

CHG
NO
01

NUMBER
0911-1.04-1226

FAILURE ANALYSIS REPORT

The broken thrust washer (the one toward the rear engine). On the broken face we found different appearance which rough dark at ¾ of the area and smooth bright at the rest. The thrust washer has cracked which then growth to cause it broken.

Sample Report

SOS report: Standard Sample Report - All Data

ITEM	ITEM NAME	QTY	UNIT	REMARK	DATE	TIME	BY	REMARK	DATE	TIME	BY	REMARK	DATE	TIME	BY	REMARK	DATE	TIME	BY	REMARK
1	THRUST WASHER	1	PC	Broken	2015	10	10	10	2015	10	10	10	2015	10	10	10	2015	10	10	10

Conclusions (What Caused the Failure): Describe the root cause and the sequence of events that caused it.

The copper found in the oil filter came from broken thrust washer P/N 7N-3006 at the PTO idler gear rear. Further inspection on broken face of thrust washer found different appearance, one side is dark rough and the other is smooth bright. The thrust washer has been cracked first before it fully broken.

The root cause most possible due to material defect on the thrust washer P/N 7N-3006

Warranty Decision or Recommendation: Please mark X in only one box below.

Not warranty :	Warranty Accepted :	ASA Claim* :
----------------	---------------------	--------------

Comments:

*) Application Severity Analysis

Solution to Prevent Reoccurrence:

Check SIS for Service Letters and Service Magazines related to this failure. AMT history, MIM's history, Oil sample results, please attach supporting documents below.

Bagian dari / Part of : TSD.TEC.SOP.001.R00
4 of 4
Form Number : PSDV.SVTC.FRM.001.R00

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Failure Analysis Report

DATE
1/13/2015CHG
NO
01NUMBER
0911-1.04-1226

3.0 Implementation Steps

- Determine the personnel who will be responsible to fill out the Failure Analysis Report form.
- Advise where the form is located in the portal or system.
- Advise all respective persons to implement using the form in the day to day process.
- Promote change feedback as installation may create new problems.
- Train all employees who have to use the new form regarding why this is required and the importance of having all completed documentation available on time at job closing.
- Review process performance once established.
- Complete any adjustment as required.

4.0 Benefits

Efficiency – With the implementation of this new format the failure analysis management will become more efficient. It will save the administration time to check that all required information is complete. If not they will also be able to see which persons is require to send the missing related information.

Increase Quality – More effective form simplifies the process - promoting high technician professionalism, higher standard and increasing brand image.

5.0 Resources Required

People – Training for those involved with Failure Analysis to advise on the importance completing the form, and the use of the new standard document.

6.0 Supporting Attachments / References

- PSDV.SVTC.FRM.001.R0 Failure Analysis Report Form

7.0 Related Best Practices

-

8.0 Acknowledgements

This Best Practice was re-written by:

- o Herwina Dewi
- o Corporate Best Practice
- o Product Support Standard and Development
- o PT. Trakindo Utama - Head Office
- o Email: herwina.dewi@trakindo.co.id
- o Office: 62217822373 ext: 1129

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Failure Analysis Report	DATE 1/13/2015	CHG NO 01	NUMBER 0911-1.04-1226
-------------------------	-------------------	-----------------	--------------------------